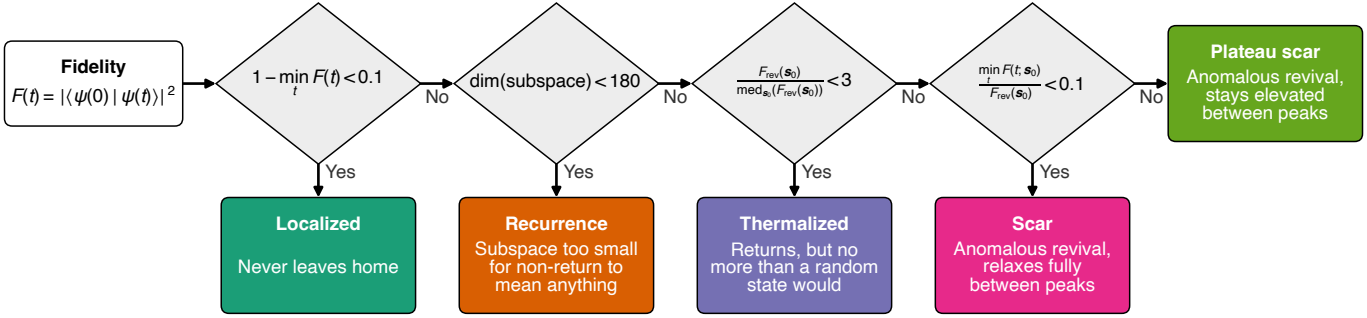
Fidelity
$F(t) = |\langle \psi(0) \mid \psi(t) \rangle|^2$
$1 - \min_t F(t) < 0.1$
No
Yes
Localized
Never leaves home
$\dim(\text{subspace}) < 180$
No
Yes
Recurrence
Subspace too small for non-return to mean anything
$\frac{F_{\text{rev}}(\boldsymbol{s}_0)}{\text{med}_{\boldsymbol{s}_0}(F_{\text{rev}}(\boldsymbol{s}_0))} < 3$
No
Yes
Thermalized
Returns, but no more than a random state would
$\frac{\min_t F(t; \boldsymbol{s}_0)}{F_{\text{rev}}(\boldsymbol{s}_0)} < 0.1$
No
Yes
Scar
Anomalous revival, relaxes fully between peaks
Plateau scar
Anomalous revival, stays elevated between peaks